

Population: population of ages 27 and older without chronic bronchitis = 95.57% of population 27+. Using the same data set, Abbey et al. (1995a, p. 140) reported that the respondents in 1977 ranged in age from 27 to 95. The American Lung Association (2010a, Table 4) reports a chronic bronchitis prevalence rate for ages 18 and over of 4.37%.

G.1.2.2 Peters et al. (2001)

See full study description under Peters et al. (2001) in Appendix E, Section E.2.5.

G.1.2.3 Pope et al. (2006)

See full study description under Pope et al. (2006) in Appendix E, Section E.2.6.

G.1.2.4 Sullivan et al. (2005)

See full study description under Sullivan et al. (2005) in Appendix E, Section E.2.9.

G.1.2.5 Zanobetti and Schwartz (2006)

See full study description under Zanobetti and Schwartz (2006) in Appendix E, Section E.2.10.

G.1.2.6 Zanobetti et al. (2009)

See full study description under Zanobetti et al. (2009) in Appendix E, Section E.2.11.

G.1.3 Hospitalizations

Table G-7 summarizes the additional health impacts functions used to estimate the relationship between PM_{2.5} and hospital admissions. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table G-7. Additional Health Impact Functions for Particulate Matter and Hospital Admissions

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Asthma	Babin et al.	2007	Washington, DC	0-17		D24HourMean	0.0020	0.0043	Log-linear	Age range adjusted Admission from emergency department only.
All Cardiovascular (less Myocardial Infarctions)	Bell et al.	2008	202 U.S. Counties	65-99		D24HourMean	0.0008	0.0001	Log-linear	Urgent admission only. Yearly national estimates
All Cardiovascular (less Myocardial Infarctions)	Bell et al.	2012	187 U.S. Counties	65-99		D24HourMean	0.0008	0.0001071	Log-linear	
Chronic Lung	Ito	2003	Detroit, MI	65-99		D24HourMean	0.001169	0.002064	Log-linear	

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Congestive Heart Failure	Ito	2003	Detroit, MI	65-99		D24HourMean	0.003074	0.001292	Log-linear	
Dysrhythmia	Ito	2003	Detroit, MI	65-99		D24HourMean	0.001249	0.002033	Log-linear	
Ischemic Heart (less Myocardial Infarctions)	Ito	2003	Detroit, MI	65-99		D24HourMean	0.001435	0.001156	Log-linear	
Pneumonia	Ito	2003	Detroit, MI	65-99		D24HourMean	0.003979	0.001659	Log-linear	
All Respiratory	Kloog et al.	2012	New England	65-99		D24HourMean	0.0007	0.0010	Log-linear	
Chronic Lung (less Asthma)	Moolgavkar	2000	Los Angeles, CA	18-64		D24HourMean	0.002200	0.000733	Log-linear	
All Cardiovascular (less Myocardial Infarctions)	Moolgavkar	2000	Los Angeles, CA	18-64		D24HourMean	0.001400	0.000341	Log-linear	
Chronic Lung	Moolgavkar	2003	Los Angeles, CA	65-99		D24HourMean	0.001850	0.000524	Log-linear	
All Cardiovascular (less Myocardial Infarctions)	Moolgavkar	2003	Los Angeles, CA	65-99		D24HourMean	0.001580	0.000344	Log-linear	
All Cardiovascular (less Myocardial Infarctions)	Peng et al.	2008	108 U.S. Counties	65-99		D24HourMean	0.0007	0.0001	Log-linear	Emergency HA
All Cardiovascular (less Myocardial Infarctions)	Peng et al.	2009	119 U.S. Counties	65-99		D24HourMean	0.0007	0.0002	Log-linear	Urgent or emergency HA
Asthma	Sheppard	2003	Seattle, WA	0-64		D24HourMean	0.003324	0.001045	Log-linear	
All Cardiovascular (less Myocardial Infarctions)	Zanobetti et al.	2009	26 U.S. Communities	65-99		D24HourMean	0.00019	0.0003	Log-linear	All seasons
All Respiratory	Zanobetti et al.	2009	26 U.S. Communities	65-99		D24HourMean	0.0021	0.0004	Log-linear	All seasons

G.1.3.1 Babin et al. (2007)

Babin et al. (2007) examined pediatric asthma-related emergency room (ER) visits and hospital admissions (ICD-9 code 493) in Washington, D.C. from 2001-2004 and their short-term associations with ozone, particulate matter, socioeconomic status, and age group. The association between PM_{2.5} and asthma hospitalization was found statistically insignificant.

Hospital Admissions, Asthma (ICD-9 code 493)

In a single-pollutant model, the coefficient and standard error are estimated from the average percent increase in risk (0.2%) and 95% confidence interval (95% CI: -0.6% - 1.1%) for a 1 µg/m³ increase in same-day daily 24-hour mean PM_{2.5} (Babin et al., 2007, Table 2).

Note that although Babin et al. (2007) reports results for the 1-17 year old age range, for comparability to other studies, we apply the results to the population of ages 0 to 17.

G.1.3.2 Bell et al. (2008, 2012)

Bell et al. (2008) evaluated the association between short-term exposure to PM_{2.5} and the risk of cardiovascular (ICD-9 codes 410-414, 26-427, 428, 429, 430-438, and 440-449) and respiratory (ICD-9 codes 464-466, 480-487, and 490-492) hospital admissions among Medicare enrollees ≥65 years old varied by season and geographic region in 202 U.S. counties with populations greater than 200,000 from 1999-2005. Three time-series models were used to provide three key variables: consistent PM effects across the year, different PM effects by season, and smoothly varying PM effects throughout the year. A two-stage Bayesian hierarchical model was used to estimate the association between PM_{2.5} and hospitalization rates, with the first stage estimating the association within a single county and the second stage combining county-specific estimates. The authors found statistically significant evidence of seasonal and regional variation. Respiratory hospitalizations were highest in winter with a 1.05% increase (95%PI: 0.29-1.82) in hospitalizations per 10 µg/m³ increase in same-day PM_{2.5}. A 1.49% increase (95% PI: 1.09-1.89) in cardiovascular hospital admissions were also found for the winter season, and associations were observed in other seasons as well. The strongest association was for the northeast for both respiratory and cardiovascular admissions.

Hospital Admissions, Cardio-, Cerebro- and Peripheral Vascular Disease (ICD-9 codes 426-427, 428, 430-438, 410-414, 429; 440-449)

For different seasons (i.e., autumn, spring, summer, winter, and all-year) and regions (i.e., southwest, northwest, southeast, southwest, and nationwide), the coefficient and standard error are estimated from the average percent increase in risk and 95% confidence interval for a 10 µg/m³ increase in same-day (lag 0) daily 24-hour mean PM_{2.5} (Bell et al., 2008, Table 2).

Note that Bell et al. (2008) considered a broader range of ICD-9 codes and estimated the risk of both cardiovascular events and cerebro- and peripheral vascular disease. For comparability to other studies, EPA decided to apply a baseline hospitalization rate for ICD-9 codes 390-409 and 411-429 when using this C-R function in quantifying impacts.

G.1.3.3 Ito (2003)

Lippmann et al. (2000) studied the association between particulate matter and daily mortality and hospitalizations among the elderly in Detroit, MI. Data were analyzed for two separate study periods, 1985-1990 and 1992-1994. The 1992-1994 study period had a greater variety of data on PM size and was the main focus of the report. The authors collected hospitalization data for a variety of cardiovascular and respiratory endpoints. They used daily air quality data for PM₁₀, PM_{2.5}, and PM_{10-2.5} in a Poisson regression model with generalized additive models (GAM) to adjust for nonlinear

relationships and temporal trends. In single pollutant models, all PM metrics were statistically significant for pneumonia (ICD codes 480-486), PM_{10-2.5} and PM₁₀ were significant for ischemic heart disease (ICD code 410-414), and PM_{2.5} and PM₁₀ were significant for heart failure (ICD code 428). There were positive, but not statistically significant associations, between the PM metrics and COPD (ICD codes 490-496) and dysrhythmia (ICD code 427). In separate co-pollutant models with PM and either ozone, SO₂, NO₂, or CO, the results were generally comparable.

In response to concerns with the Splus issue, Ito (2003) reanalyzed the study by Lippmann et al. (2000). The reanalysis by Ito reported that more generalized additive models with stringent convergence criteria and generalized linear models resulted in smaller relative risk estimates.

Chronic Lung Disease (ICD-9 codes 490-496)

The coefficient and standard error are based on the relative risk (1.043) and 95% confidence interval (0.902-1.207) for a 36 µg/m³ increase in PM_{2.5} in the 3-day lag GAM stringent model (Ito, 2003, Table 8).

Congestive Heart Failure (ICD-9 code 428)

The co-pollutant coefficient and standard error are calculated from a relative risk of 1.117 (95% CI 1.020-1.224) for a 36 µg/m³ increase in PM_{2.5} in the 1-day lag GAM stringent model (Ito, 2003, Table 11).

Dysrhythmia (ICD-9 code 427)

The co-pollutant coefficient and standard error are calculated from a relative risk of 1.046 (95% CI 0.906-1.207) for a 36 µg/m³ increase in PM_{2.5} in the 1-day lag GAM stringent model (Ito, 2003, Table 10).

Ischemic Heart Disease (ICD-9 codes 411-414)

The co-pollutant coefficient and standard error are calculated from a relative risk of 1.053 (95% CI 0.971-1.143) for a 36 µg/m³ increase in PM_{2.5} in the 1-day lag GAM stringent model (Ito, 2003, Table 9). Note that Lippmann et al. (2000) report results for ICD codes 410-414. In the benefit analysis, avoided nonfatal heart attacks are estimated using the results reported by Peters et al. (2001). The baseline rate in the Peters et al. function is a modified heart attack hospitalization rate (ICD code 410), since most, if not all, nonfatal heart attacks will require hospitalization. In order to avoid double counting heart attack hospitalizations, we have excluded ICD code 410 from the baseline incidence rate used in this function.